

Proxmox VE Cheat Sheet (qm & pct)

Manage Proxmox VMs (qm) and LXC containers (pct) from the host's shell, as root. Web UI: <https://192.168.1.10:8006>

Using it

qm COMMAND 100 ← VMs are addressed by number (VMID), not name
pct COMMAND 101 ← same idea for containers; VMs and containers share one set of IDs

Reading qm list

VMID	NAME	STATUS	MEM(MB)	BOOTDISK(GB)	PID	Column	Means
100	web1	running	4096	32.00	2817	VMID	Number used in commands
101	db1	running	8192	64.00	3105	MEM(MB)	RAM assigned (not used)
102	win-test	stopped	8192	80.00	0	BOOTDISK(GB)	Size of its boot disk
						PID	Host process ID; 0 = off

VMs (qm ...)

Command	Does	Command	Does
start 100	Boots it	snapshot 100 snap1	Snapshot named snap1
shutdown 100	Asks the guest to shut down	listsnapshot 100	Lists its snapshots
stop 100	Pulls the plug	rollback 100 snap1	Goes back to snap1
reboot 100	Clean reboot	clone 100 120 --full	Copy to new VMID 120
status 100	running / stopped	terminal 100	Serial console; exit Ctrl-C
config 100	Shows its settings	monitor 100	QEMU monitor (advanced)
set 100 --memory 8192	RAM in MB (--cores 4)	unlock 100	Clears a stuck "VM is locked"
set 100 --onboot 1	Start when the host boots	destroy 100	Deletes it and its disks

Containers (pct ...)

Command	Does	Command	Does
list	All containers and their state	enter 101	Root shell inside it
start / shutdown 101	Start / clean stop	exec 101 -- df -h	Runs one command inside
config 101	Shows its settings	set 101 --cores 2	Same options as qm set
snapshot 101 snap1	Snapshot (also rollback)	destroy 101	Deletes it and its disk

The host

Command	Does
pveversion -v	Proxmox and package versions
pvesm status / pvecm status	Storage space / cluster nodes and quorum
vzdump 100 --mode snapshot --compress zstd	Backs up 100 while it runs (to local storage)
qmrestore /var/lib/vz/dump/vzdump-qemu-100-....vma.zst 120	Restores a VM backup as 120
pct restore 121 /var/lib/vz/dump/vzdump-lxc-101-....tar.zst	Restores a container backup as 121
pveam update / pveam available --section system	Refresh / list container templates
pveam download local debian-12-standard_....tar.zst	Download one (name from the list)

The bridge (vbr0)

```
iface eno1 inet manual ← the NIC: no IP of its own
auto vbr0 ← bring up at boot
iface vbr0 inet static
    address 192.168.1.10/24 ← the host's IP
    gateway 192.168.1.1
    bridge-ports eno1 ← NIC plugged in
    bridge-stp off
    bridge-fd 0
```

/etc/network/interfaces: vbr0 does the job main does on a plain host (the web UI expects vbrN names).

ifreload -a applies edits without a reboot.

Put a VM on it: qm set 100 --net0 virtio,bridge=vbr0

Web UI edits wait in interfaces.new until Apply Configuration.

Remember: update with **apt update && apt dist-upgrade**, never plain **apt upgrade**. "VM is locked" after a failed backup? **qm unlock 100**.